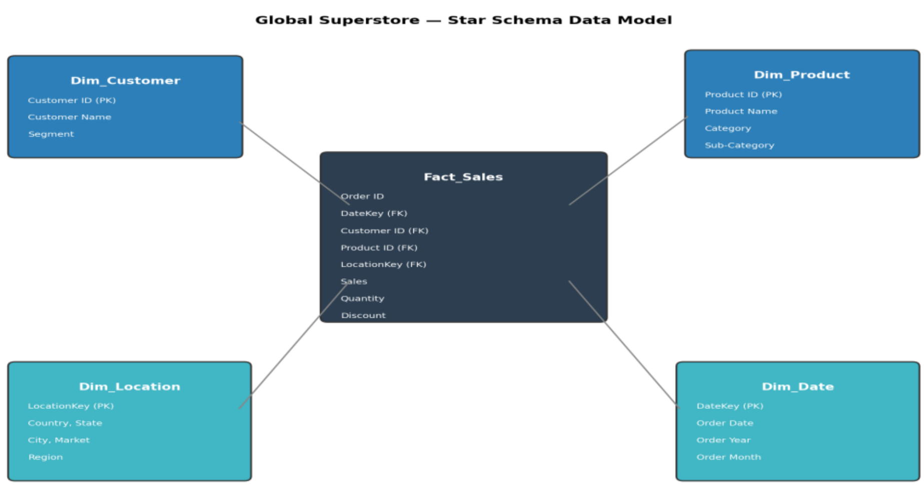
GLOBAL SUPERSTORE

Data Cleaning, Dimensional Data Warehousing, SQL Analysis & Power BI Dashboarding



Submitted by

1. Osama Medhat Elsayed

2. Adham Mohamed Abdelaziz

3.Nada Mostafa Atiyah

4. Habiba Hany Mostafa

Submitted to

Prof.Dr : Muhammad Safwat

Eng : Dina Mohsen

AUG 2026

Table of Contents

1. Executive Summary

This document presents the complete methodology, technical implementation, and findings of a full-cycle data analytics project built on the Global Superstore dataset — a retail transaction dataset spanning four years (2011-2014), 51,290 order line items, 147 countries, and three product categories. The project was designed to simulate a real-world business-intelligence workflow, taking raw transactional data through cleaning, dimensional modeling, SQL-based analysis, and interactive dashboarding.

The workflow was executed in five phases: (1) data cleaning and feature engineering in Python (Pandas), (2) exploratory data analysis and visualization, (3) construction of a star-schema data warehouse (one fact table and four dimension tables), (4) SQL analysis across three business-focused query sets, and (5) an interactive five-page Power BI report connecting the warehouse tables to executive, product, and customer/geographic views.

Across the full dataset, the business generated approximately $12.64 million in total sales and $1.47 million in net profit across 25,035 distinct orders, at an average sale value of $246.49 per line item. The Consumer segment, the APAC market, and the Technology category emerged as the strongest overall performers, while high discount levels (above 20%) were shown to erode — and in many cases reverse — profitability

This report is organized to mirror the actual project pipeline: raw data → cleaned data → dimensional warehouse → SQL analysis → Power BI dashboard → business insights and recommendations. Each phase includes the exact steps performed, the reasoning behind key decisions, and the visual or tabular evidence produced at that stage.

2. Introduction

2.1 Project Background

Retail organizations generate large volumes of transactional data every day — orders, shipments, discounts, and returns spread across countries, product lines, and customer segments. Raw transactional data of this kind is rarely analysis-ready: it contains redundant columns, inconsistent date formats, and no structure that supports fast aggregation across business questions. This project takes on the role of a data analyst tasked with turning such raw data into a governed, query-ready warehouse and a set of decision-support dashboards for company leadership.

2.2 Objectives

- Clean and validate the raw Global Superstore dataset, resolving data-quality issues and engineering analysis-ready features.
- Explore the data visually to understand distributions, relationships, and early business signals before formal modeling.
- Design and build a dimensional (star-schema) data warehouse to support fast, reusable SQL analysis.
- Write structured SQL queries to answer specific business questions across three functional areas: executive performance, product performance, and customer/geographic performance.
- Design an interactive Power BI report that translates the SQL analysis into a self-service, filterable dashboard for stakeholders.
- Summarize the findings into clear, actionable business insights and recommendations.

2.3 Tools & Technologies

Tool	Role in the Project
Python (Pandas, NumPy)	Data cleaning, validation, and feature engineering
Matplotlib / Seaborn / Plotly	Exploratory data analysis and static visualizations
Google Colab	Cloud notebook environment used to run the cleaning and warehouse-build notebooks
SQL Server (T-SQL)	Dimensional schema querying — KPI, ranking, and grouping analysis
Power BI Desktop	Interactive, multi-page dashboard built on the star-schema tables
Microsoft Excel / CSV	Interchange format for moving cleaned and dimensional tables between tools

3. Dataset Overview

3.1 Data Source

The analysis is based on the Global Superstore dataset (Global_Superstore2.csv), a widely used retail-sales benchmark dataset. In its raw form the dataset contains 51,290 rows and 24 columns, covering order-level transactions placed between 2011 and 2014 across seven markets (APAC, EU, US, LATAM, EMEA, Africa, and Canada), 147 countries, and 3 top-level product categories.

3.2 Raw Structure at a Glance

Property	Value
Rows (order line items)	51,290
Columns (raw)	24
Distinct orders	25,035
Date range	2011 – 2014
Countries / Cities	147 / 3,636
Markets / Regions	7 / 13
Customers	1,590
Products / Categories / Sub-categories	10,292 / 3 / 17
Duplicate rows found	0
Fully-null column found	Postal Code (only 9,994 of 51,290 rows populated)

3.3 Data Dictionary (Key Fields)

Field	Type	Description
Order ID / Row ID	Text / Int	Order-level and line-item identifiers
Order Date / Ship Date	Date	Order placement and shipment dates
Ship Mode	Text	Standard Class, Second Class, First Class, Same Day
Customer ID / Name / Segment	Text	Customer identity and segment: Consumer, Corporate, Home Office
City / State / Country / Market / Region	Text	Geographic hierarchy of the order
Product ID / Name / Category / Sub-Category	Text	Product hierarchy
Sales / Quantity / Discount / Profit	Numeric	Transaction-level order value, units, discount rate, and profit
Shipping Cost / Order Priority	Numeric / Text	Freight cost and priority flag (Low/Medium/High/Critical)

4. Phase 1 — Data Cleaning & Preparation

Data cleaning was carried out in Python inside a Google Colab notebook (final_project_clean.ipynb). The dataset was loaded directly from Google Drive with pd.read_csv(..., encoding="latin1") to correctly handle the accented characters present in international city and customer names.

4.1 Initial Exploration

The dataset was first inspected using df.head(), df.tail(), df.shape, df.info(), df.dtypes, and df.describe() (for both numeric and object columns). This step confirmed 51,290 rows and 24 columns, and surfaced that all date columns and the Postal Code field were loaded as generic object/float types rather than proper date or nullable types — the first cleaning targets.

4.2 Missing Value Assessment

df.isnull().sum() showed that every column was fully populated except Postal Code, which had 41,296 missing values out of 51,290 rows (81% missing). Because Postal Code was not required for any planned geographic or business analysis — City, State, Country, Market, and Region already provide a full location hierarchy — the column was dropped rather than imputed, avoiding the introduction of artificial values into the dataset.

4.3 Duplicate & Validity Checks

Two categories of checks were performed to confirm the dataset was analysis-ready:

- Exact duplicate rows: df.duplicated().sum() returned 0 — no duplicate records were present.
- Business-rule validation: Quantity ≤ 0 → 0 rows; Discount outside the 0–1 range → 0 rows; Sales < 0 → 0 rows; Profit < 0 → 12,544 rows (23.9% of all rows).

Negative profit is expected in retail data — it reflects orders sold at a loss (typically after heavy discounting) rather than a data-quality defect, so these rows were retained and instead used later as an input to the discount-vs-profitability analysis .

4.4 Data Type Conversion

Order Date and Ship Date were converted from text to proper datetime64 objects using pd.to_datetime(..., dayfirst=True), matching the dataset's day/month/year text format. This conversion is what enables all downstream date arithmetic (shipping-time calculation) and the Order Year / Order Month features engineered in the next step.

4.5 Feature Engineering

Four new analytical fields were engineered directly from the cleaned columns:

New Field	Logic	Purpose
Order Year	Order Date.dt.year	Enables year-over-year trend analysis
Order Month	Order Date.dt.month_name()	Enables seasonal / monthly trend analysis
Shipping Days	(Ship Date − Order Date).days	Measures fulfilment speed
Discount Level	No Discount (0) · Low (≤20%) · Medium (≤50%) · High (>50%)	Buckets discount rate for profitability analysis
Profit Category	Loss (<0) · Low (<$100) · Medium (<$500) · High (≥$500)	Buckets order profitability for segmentation

4.6 Frequency Analysis

Before visualizing, value_counts() was run on the key categorical fields to understand the shape of the data:

Field	Distribution
Category	Office Supplies 31,273 · Technology 10,141 · Furniture 9,876
Segment	Consumer 26,518 · Corporate 15,429 · Home Office 9,343
Ship Mode	Standard Class 30,775 · Second Class 10,309 · First Class 7,505 · Same Day 2,701
Market	APAC 11,002 · LATAM 10,294 · EU 10,000 · US 9,994 · EMEA 5,029 · Africa 4,587 · Canada 384
Order Priority	Medium 29,433 · High 15,501 · Critical 3,932 · Low 2,424

The cleaned, feature-engineered dataset was exported as Global_Superstore_Cleaned.csv and used as the single source of truth for every downstream stage of the project — the data warehouse build, the SQL dashboards, and the Power BI report.

5. Phase 2 — Exploratory Data Analysis & Visualization

Before formal SQL analysis, the cleaned dataset was explored visually using Matplotlib and Seaborn to sanity-check the frequency analysis and surface early business signals. Four charts were produced directly in the cleaning notebook.

5.1 Sales by Category

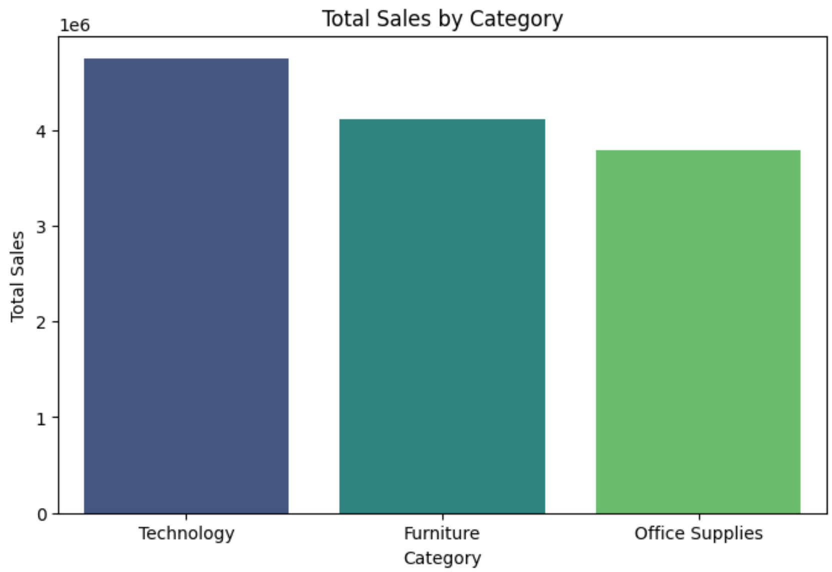


Figure 5.1 — Total sales by product category.

Office Supplies leads in order volume but, as later confirmed in the SQL analysis, Technology generates the highest sales and profit per order — a first hint that volume and value do not move together in this dataset.

5.2 Profit by Market

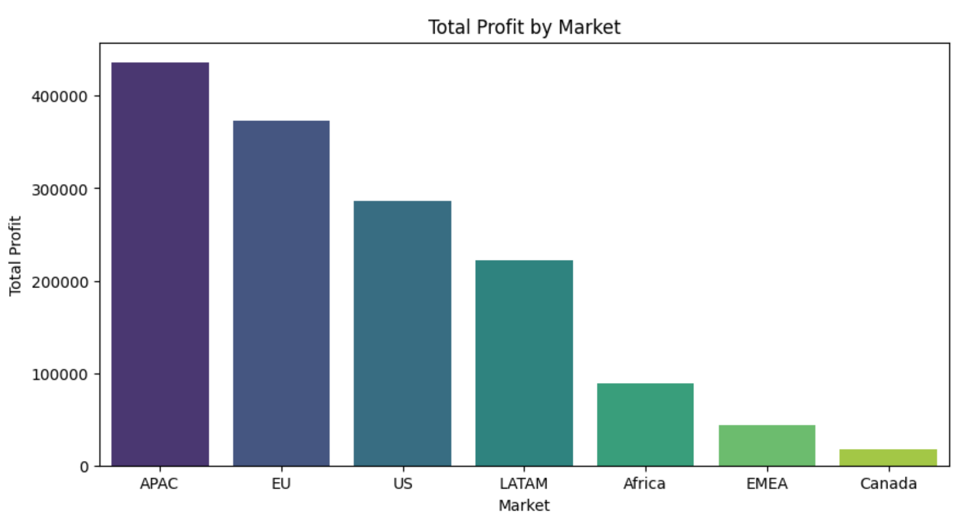


Figure 5.2 — Total profit by market.

5.3 Customer Segment Distribution

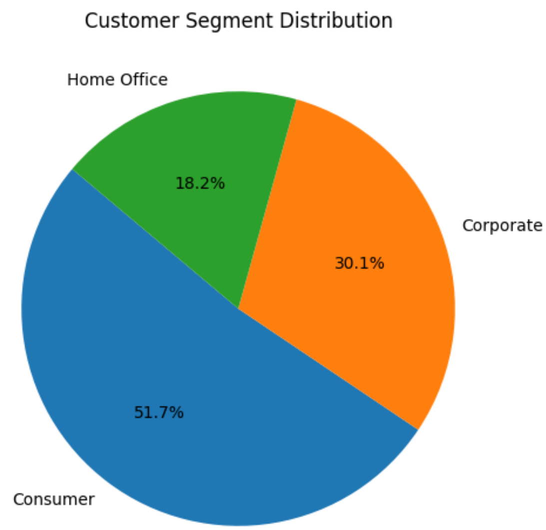


Figure 5.3 — Share of orders by customer segment.

The Consumer segment accounts for roughly half of all orders, making it the natural focus of any segment-level retention or promotion strategy.

5.4 Monthly Sales Trend

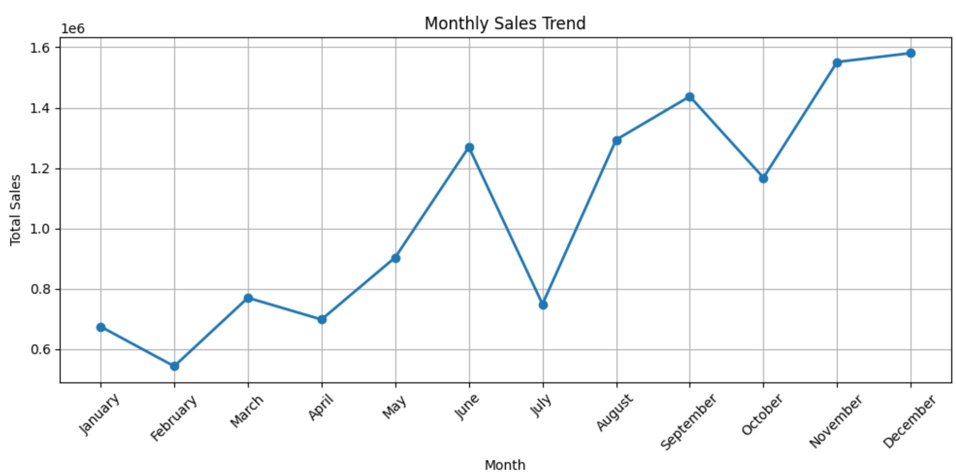


Figure 5.4 — Total sales aggregated by calendar month, across all years.

Sales rise steadily from a low point early in the year toward a strong Q4 peak (November–December), a classic seasonal retail pattern that has direct implications for inventory planning and staffing.

6. Phase 3 — Data Warehouse Design (Star Schema)

To support fast, reusable SQL analysis, the cleaned flat file was restructured into a star schema — one central fact table surrounded by four dimension tables. This normalization removes repeated descriptive data (customer names, product names, city/country text) from the transactional grain and replaces it with compact keys, which is both more storage-efficient and easier to query and filter in SQL Server and Power BI.

6.1 Schema Diagram

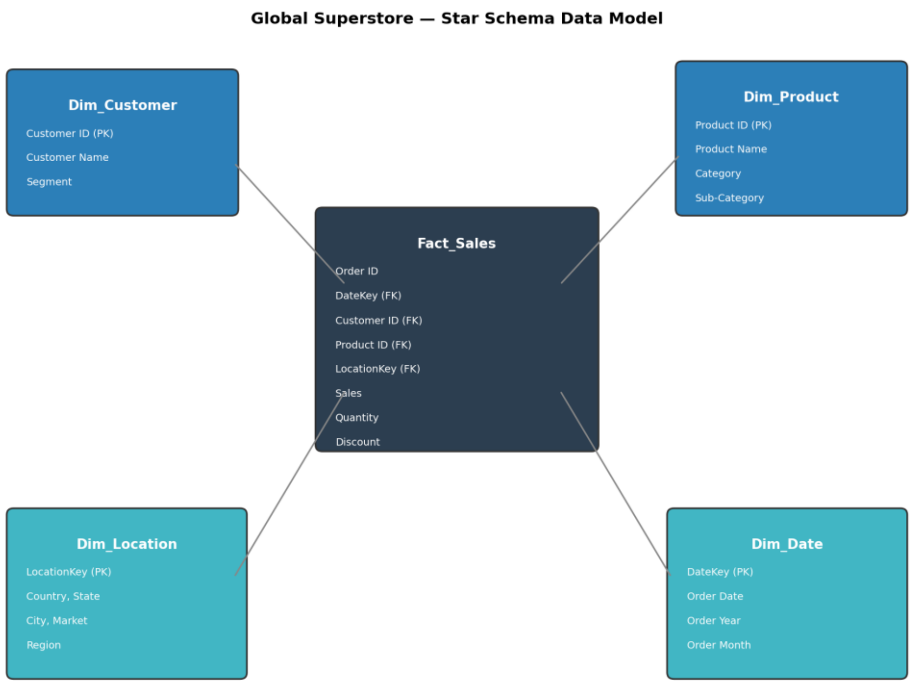


Figure 6.1 — Star schema: Fact_Sales at the center, joined to four dimension tables.

6.2 Dimension Tables

Table	Grain / Key	Columns
Dim_Customer	Customer ID (natural key)	Customer ID, Customer Name, Segment
Dim_Product	Product ID (natural key)	Product ID, Product Name, Category, Sub-Category
Dim_Location	LocationKey (surrogate key, generated)	LocationKey, Country, State, City, Market, Region
Dim_Date	DateKey (surrogate key, generated)	DateKey, Order Date, Order Year, Order Month

6.3 Fact Table

Fact_Sales holds one row per order line item — the same grain as the cleaned source file — with the descriptive columns replaced by foreign keys: Order ID, DateKey, Customer ID, Product ID, LocationKey, plus the numeric measures Sales, Quantity, Discount, Profit, and Shipping Cost.

6.4 Build Process (ETL Logic)

The warehouse was built in a second notebook (final_project_warehouse.ipynb) with the following steps, executed against the cleaned CSV:

1. Load Global_Superstore_Cleaned.csv from Google Drive.
2. Build Dim_Product, Dim_Customer by selecting and drop_duplicates() on their respective natural-key columns.
3. Build Dim_Location and Dim_Date by selecting and de-duplicating their columns, then assigning a sequential surrogate key (LocationKey, DateKey) with range(1, len(dim)+1).
4. Merge the surrogate keys back onto the main dataframe with left joins on the natural columns (Country/State/City/Market/Region for location; Order Date for date).
5. Select the fact-table columns (IDs, keys, and the five numeric measures) into Fact_Sales.
6. Export all five tables to CSV (Dim_Product.csv, Dim_Customer.csv, Dim_Location.csv, Dim_Date.csv, Fact_Sales.csv) for loading into SQL Server and Power BI.

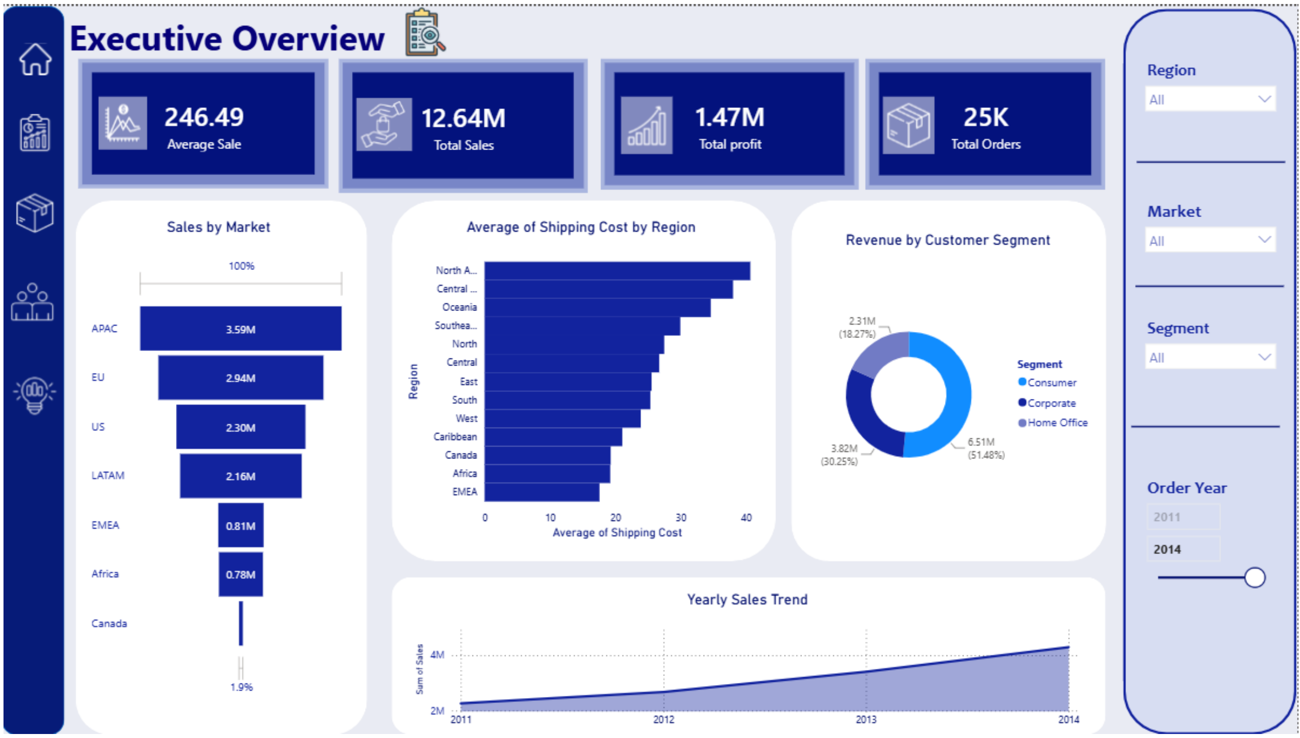
This ETL process was validated by re-aggregating Fact_Sales after the joins: the row count remained 51,290 (matching the cleaned source with no join fan-out), confirming that both dimension tables were properly de-duplicated on their join keys before the merge.

7. Phase 4 — SQL Analysis

With the star schema loaded into SQL Server, three T-SQL query sets were written, each corresponding to one business-facing dashboard and matching the structure later built in Power BI. Every query joins Fact_Sales to the relevant dimension table(s) on the surrogate or natural keys .

7.1 Dashboard 1 — Executive Overview

This dashboard answers top-level performance questions for company leadership: overall KPIs, market comparison, discount impact, shipping cost, segment revenue, and the yearly trend.




```
-- Total Revenue (KPI)
SELECT ROUND(SUM(Sales),2) AS Total_Sales FROM Fact_Sales;

-- Which market generates the highest sales?
SELECT l.Market, ROUND(SUM(f.Sales),2) AS Total_Sales
FROM Fact_Sales f
JOIN Dim_Location l ON f.LocationKey = l.LocationKey
GROUP BY l.Market
ORDER BY Total_Sales DESC;

-- Does increasing discount reduce profit?
SELECT ROUND(Discount,2) AS Discount,
       ROUND(SUM(Sales),2) AS Total_Sales,
       ROUND(SUM(Profit),2) AS Total_Profit
FROM Fact_Sales
GROUP BY Discount
ORDER BY Discount;
```

Executive KPIs (verified against the source data)

KPI	Value
Total Sales	$12,642,501.91
Total Profit	$1,467,457.29
Total Orders	25,035
Average Sale (per line item)	$246.49

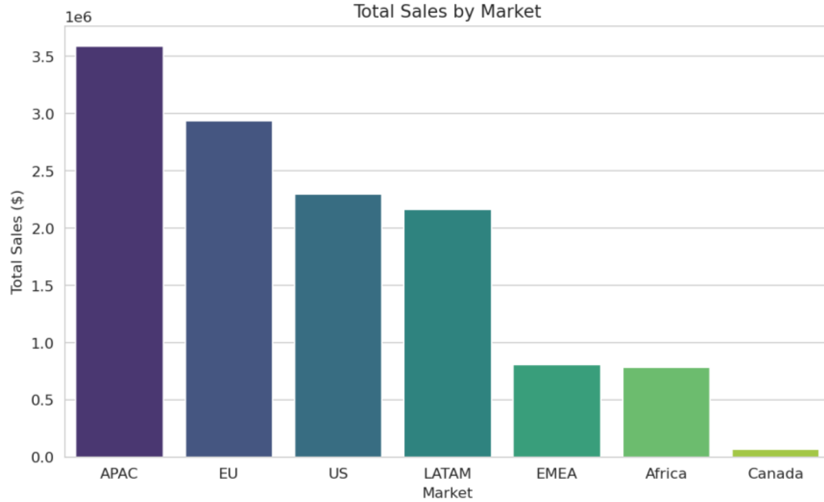


Figure 7.1 — Total sales by market. APAC leads, followed by EU and the US.

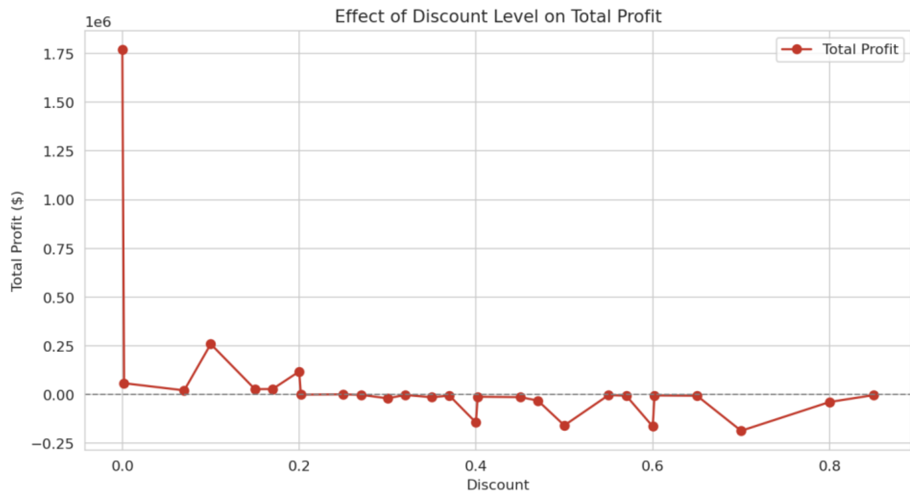
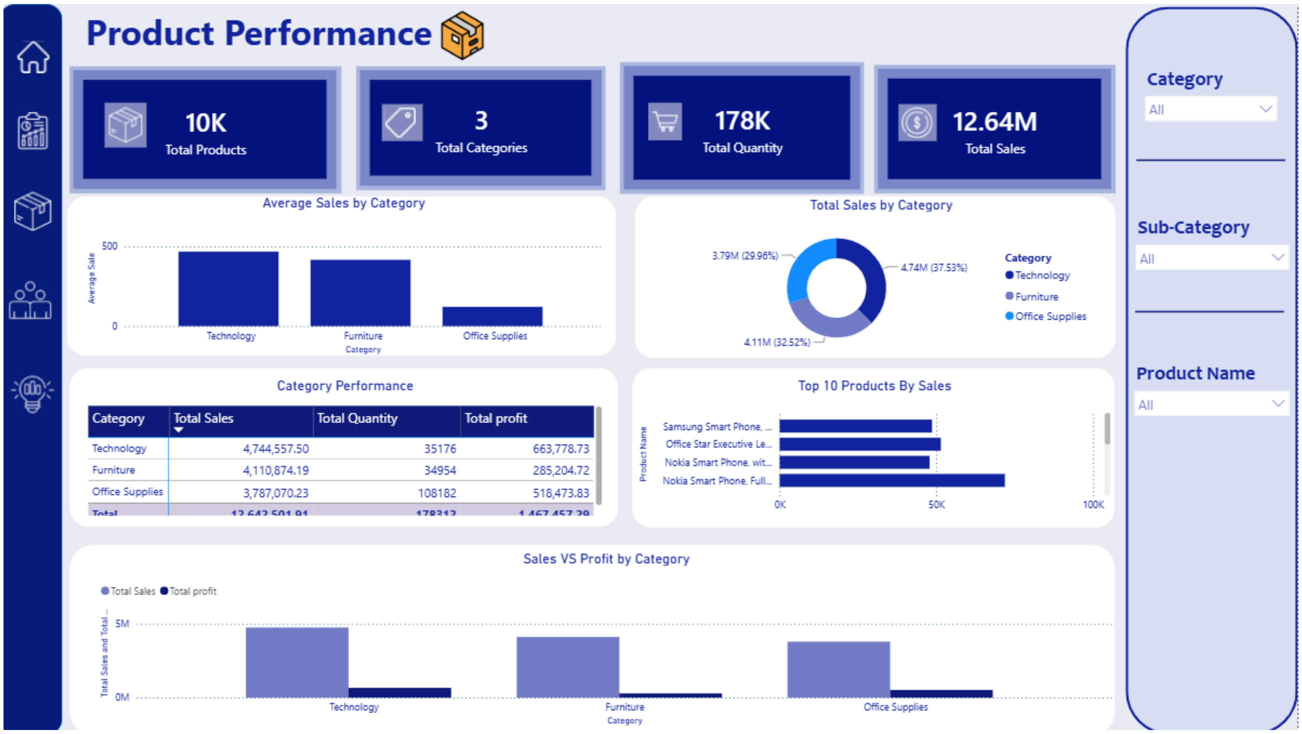


Figure 7.2 — Total profit collapses and turns negative once discount rates pass roughly 25–30%.

7.2 Dashboard 2 — Product Performance

This dashboard evaluates the product catalogue: category-level performance, top sellers, and profitability by category.




```
-- Product Category Performance
SELECT p.Category,
       ROUND(SUM(f.Sales),2)  AS Total_Sales,
       SUM(f.Quantity)        AS Total_Quantity,
       ROUND(SUM(f.Profit),2) AS Total_Profit
FROM Fact_Sales f
JOIN Dim_Product p ON f.Product_ID = p.Product_ID
GROUP BY p.Category
ORDER BY Total_Sales DESC;

-- Top 10 products by sales
SELECT TOP 10 p.Product_Name, ROUND(SUM(f.Sales),2) AS Total_Sales
FROM Fact_Sales f
JOIN Dim_Product p ON f.Product_ID = p.Product_ID
GROUP BY p.Product_Name
ORDER BY Total_Sales DESC;
```

Category	Total Sales	Total Quantity	Total Profit
Technology	$4,744,557	35,176	$663,779
Furniture	$4,110,874	34,954	$285,205
Office Supplies	$3,787,070	108,182	$518,474

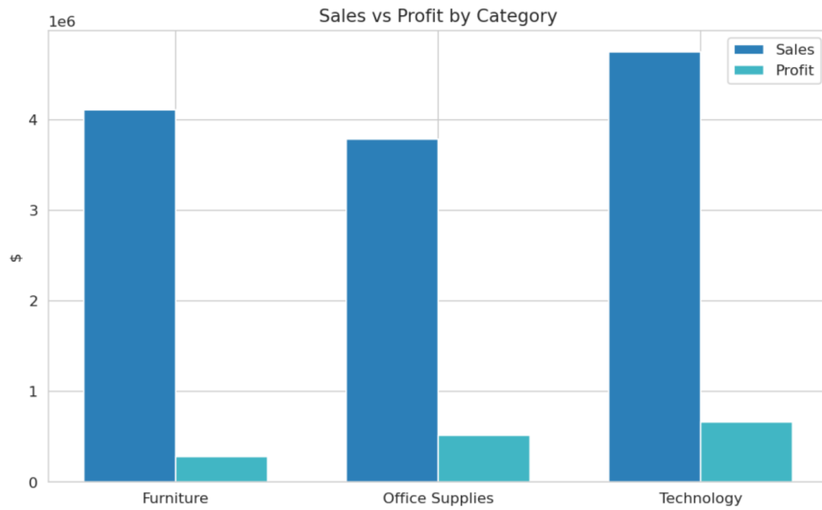


Figure 7.3 — Sales vs. profit by category. Technology is the smallest by volume but the strongest earner.

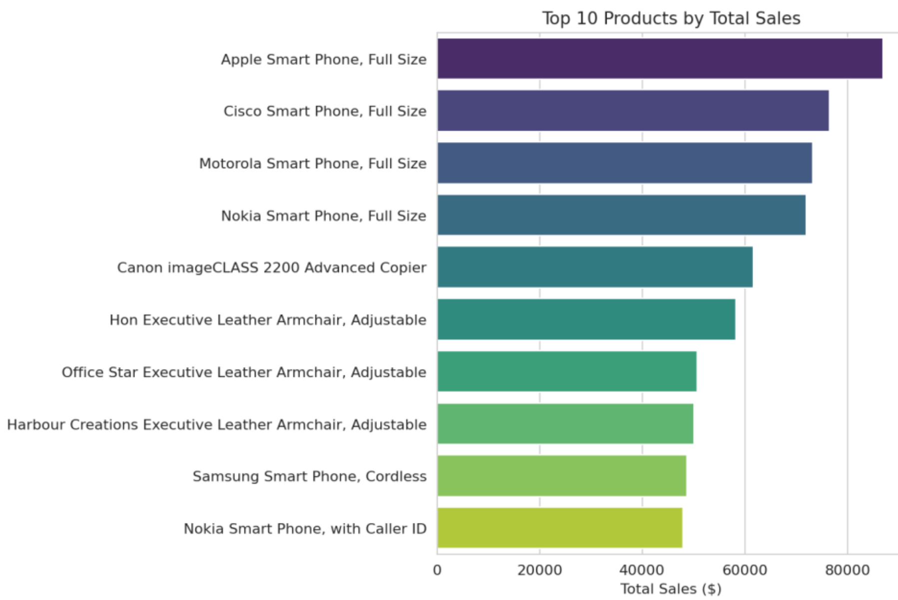
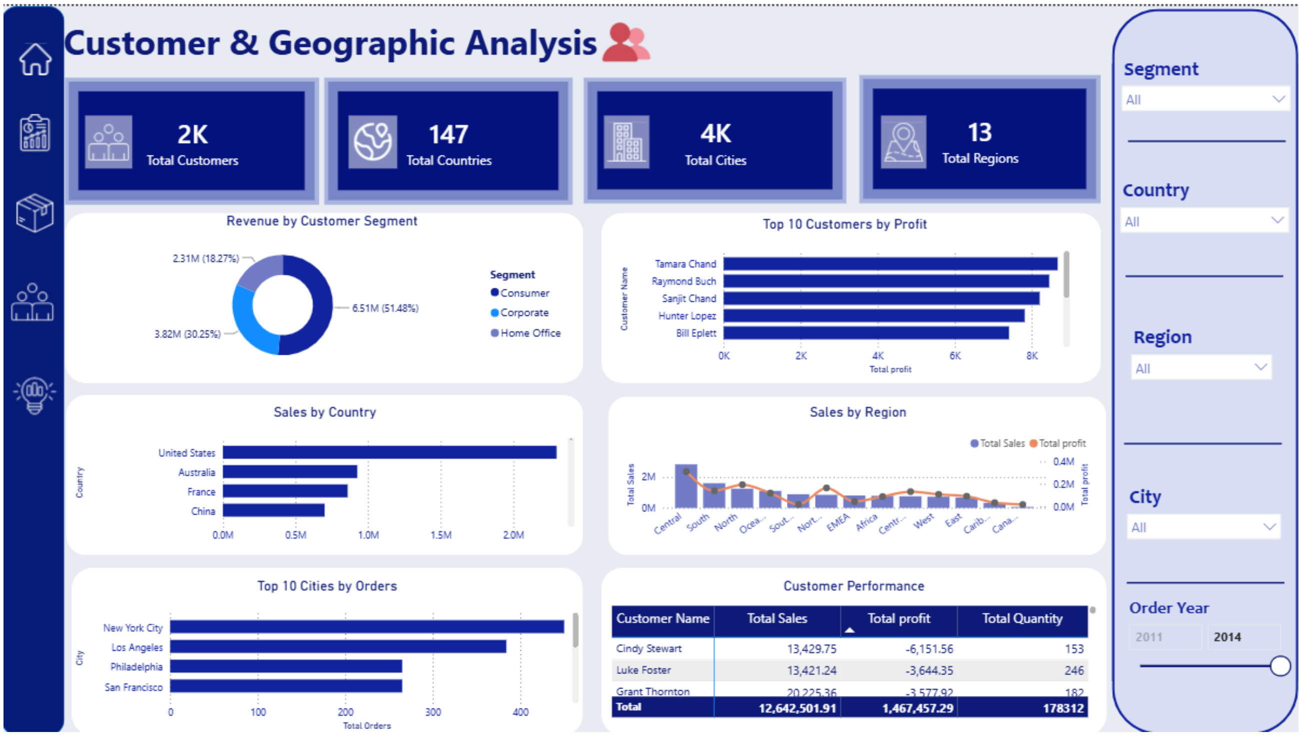


Figure 7.4 — Top 10 products by total sales; smartphones dominate the list.

7.3 Dashboard 3 — Customer & Geographic Analysis

This dashboard shifts focus to who is buying and where: segment revenue, top customers, and performance by country, region, and city.




```
-- Top 10 Customers by Profit
SELECT TOP 10 c.Customer_Name, ROUND(SUM(f.Profit),2) AS Total_Profit
FROM Fact_Sales f
JOIN Dim_Customer c ON f.Customer_ID = c.Customer_ID
GROUP BY c.Customer_Name
ORDER BY Total_Profit DESC;

-- Sales by Country
SELECT l.Country, ROUND(SUM(f.Sales),2) AS Total_Sales
FROM Fact_Sales f
JOIN Dim_Location l ON f.LocationKey = l.LocationKey
GROUP BY l.Country
ORDER BY Total_Sales DESC;
```

KPI	Value
Total Customers	1,590
Total Countries	147
Total Cities	3,636
Total Regions	13

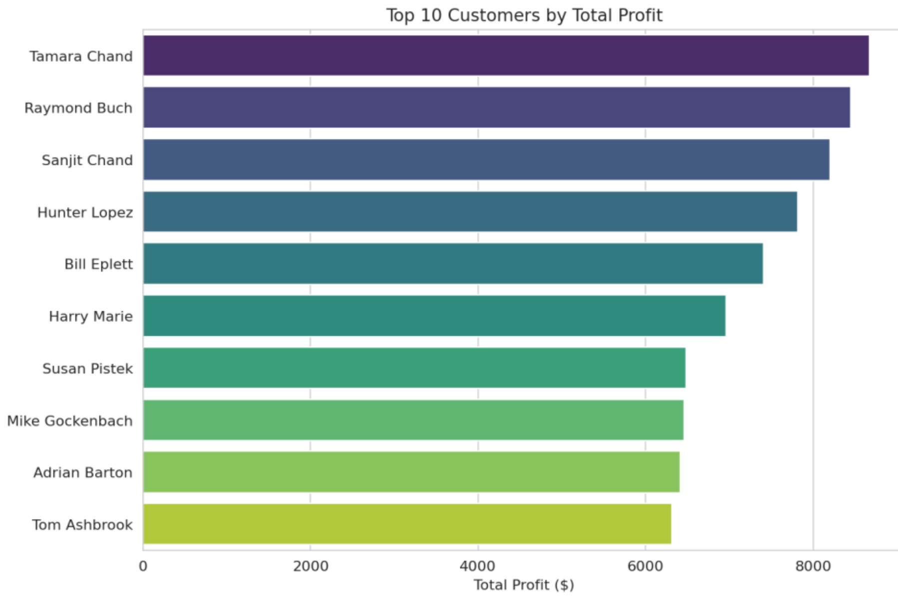


Figure 7.5 — Top 10 customers by total profit.

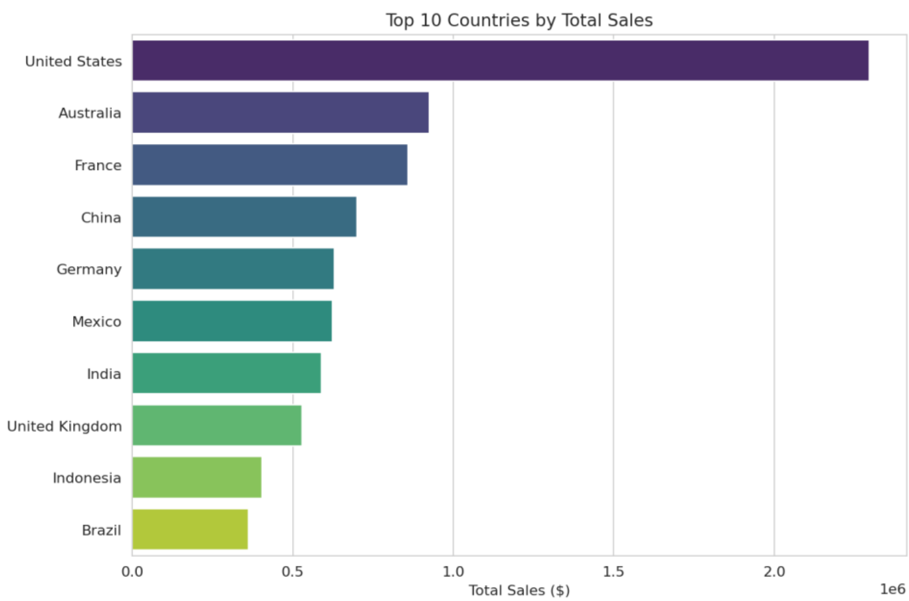


Figure 7.6 — Top 10 countries by total sales; the United States leads by a wide margin.

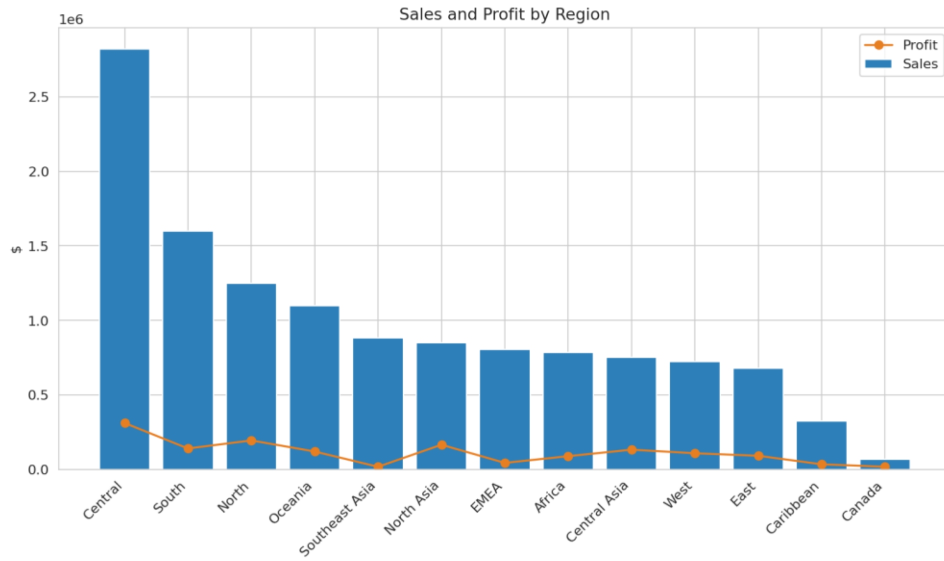


Figure 7.7 — Sales (bars) and profit (line) by region.

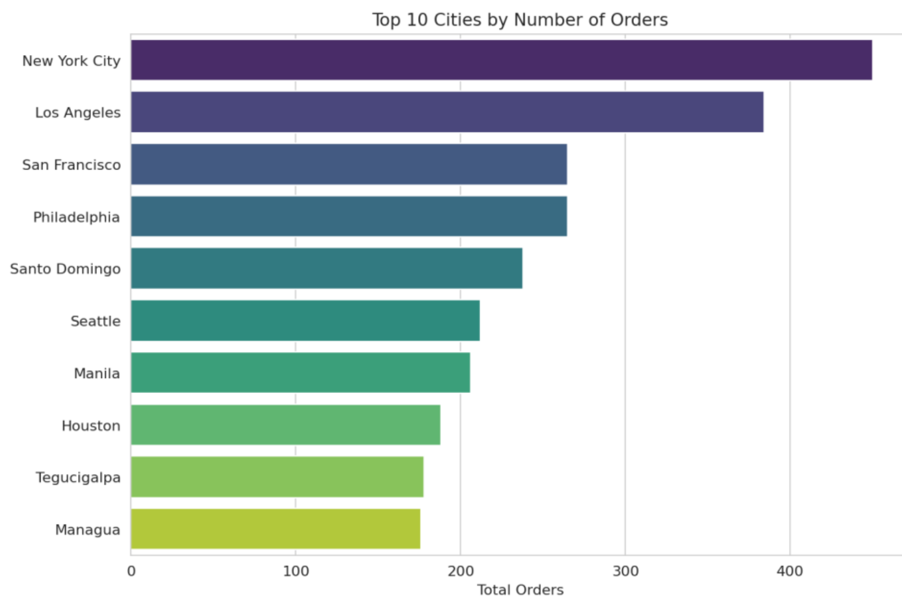


Figure 7.8 — Top 10 cities by number of distinct orders.

8. Phase 5 — Power BI Dashboard (Three Dashboards)

The five Fact/Dimension CSV exports were loaded into Power BI Desktop (finall.pbix) and modeled using the same star-schema relationships. The resulting report contains five pages, mirroring the cover, the three SQL dashboards, and a summary insights page.

8.1 Report Structure

Page	Purpose
Cover	Title page and navigation entry point for the report
Executive Overview	KPI cards, sales-by-market funnel, revenue-by-segment donut, and yearly sales trend line
Product Performance	KPI cards, average sales by category, category sales donut, product performance pivot table, and top 10 products bar chart
Customer & Geographic Analysis	KPI cards, revenue-by-segment donut, top customers and top cities bar charts, sales-by-country bar chart, and a sales-by-region combo chart
Key Business Insights	Narrative summary page consolidating the findings from all three analytical pages

8.2 Page-by-Page Visual Inventory

The table below lists every data visual placed on each report page, extracted directly from the Power BI report layout definition.

Page	Visual Type	Title
Executive Overview	Card × 4	Total Sales, Total Profit, Total Orders, Average Sale
Executive Overview	Funnel chart	Sales by Market
Executive Overview	Clustered bar chart	Average Shipping Cost by Region
Executive Overview	Donut chart	Revenue by Customer Segment
Executive Overview	Line chart	Yearly Sales Trend
Product Performance	Card × 4	Total Products, Total Categories, Total Sales, Total Quantity
Product Performance	Column chart	Average Sales by Category
Product Performance	Donut chart	Total Sales by Category
Product Performance	Pivot table	Category Performance
Product Performance	Clustered bar chart	Top 10 Products by Sales
Product Performance	Clustered column chart	Sales vs Profit by Category
Customer & Geographic Analysis	Card × 4	Total Customers, Total Countries, Total Cities, Total Regions
Customer & Geographic Analysis	Donut chart	Revenue by Customer Segment

Page	Visual Type	Title
Customer & Geographic Analysis	Bar chart	Top 10 Customers by Profit
Customer & Geographic Analysis	Bar chart	Sales by Country
Customer & Geographic Analysis	Line + stacked column combo	Sales by Region
Customer & Geographic Analysis	Bar chart	Top 10 Cities by Orders
Customer & Geographic Analysis	Pivot table	Customer Performance

Each analytical page also includes slicers (segment, category, market/region, and year, depending on the page) so end users can filter every visual on the page interactively, and navigation buttons that let a viewer jump directly between the Cover, Executive Overview, Product Performance, Customer & Geographic Analysis, and Key Business Insights pages without using the page tabs.

9. Key Business Insights

Bringing together the EDA, SQL analysis, and Power BI report, the following insights summarize what the data shows about the business:

9.1 Overall Performance

Key Insights

- Total sales reached $12.64 million and total profit reached $1.47 million across 25,035 distinct orders, for an average sale of $246.49 per line item.
- Sales grew every year in the dataset — from $2.26M in 2011 to $4.30M in 2014 — a strong, consistent upward trend.

9.2 Discounting & Profitability

Key Insights

- At 0% discount, the business earned $1.77M in profit on $6.99M of sales — a healthy ~25% margin.
- Once discount rates exceed roughly 20–30%, cumulative profit turns negative: sales in the 21–50% discount band generated a net loss of over $400,000.
- This is one of the clearest, most actionable patterns in the dataset: aggressive discounting is directly associated with unprofitable orders.

9.3 Product & Category Performance

Key Insights

- Technology is the strongest category by both total profit ($663,779) and average profit per order ($65.45), despite Office Supplies generating three times the order volume.
- The best-selling individual products are smartphones (Apple, Cisco, Motorola, and Nokia models) and executive leather armchairs.
- Office Supplies drives the highest unit volume (108,182 units) but the lowest average sale and profit per order — a high-volume, low-margin category.

9.4 Customers & Geography

Key Insights

- The Consumer segment is the largest revenue contributor ($6.51M) and the largest order volume segment (26,518 orders, ~52% of all orders).

- The United States is the single largest country by sales ($2.30M), followed by Australia and France.
- The Central region leads all 13 regions in both total sales ($2.82M) and total profit ($311K).
- A small group of top customers (e.g. Tamara Chand, Raymond Buch, Sanjit Chand) each contribute over $8,000 in cumulative profit, underlining the value of a formal customer-retention or loyalty strategy.

9.5 Operations

Key Insights

- North Asia and Central Asia carry the highest average shipping cost per order (>$38), more than double the cost seen in EMEA and Africa — a logistics cost worth investigating further.
- Standard Class is used for 60% of all shipments, suggesting most customers are not paying for expedited delivery.

10. Recommendations

the following actions are recommended:

1. Cap discretionary discounting near 20%. Order-level data shows profitability turns negative well before the 50% discount mark; a formal discount-approval threshold around 20% would protect margin on the majority of transactions.
2. Double down on Technology. It already delivers the best margin per order; expanding assortment or promoting Technology cross-sells to Consumer-segment shoppers is likely to be margin-accretive.
3. Re-evaluate Furniture pricing and cost structure. Furniture generates nearly as much revenue as Technology but less than half the profit — a strong candidate for a supplier-cost or pricing review.
4. Investigate North Asia / Central Asia shipping costs. These regions' shipping costs are more than double the company average; consolidating carriers or warehousing locally could meaningfully improve regional margin.
5. Build a formal loyalty program for top-decile customers. A small number of customers already drive a disproportionate share of profit; a retention program targeted at this group protects high-margin revenue at low incremental cost.
6. Use the Power BI report as a recurring monthly review tool. The slicers on Year, Segment, Category, and Region make the existing report suitable for a standing monthly business review rather than a one-time analysis.

11. Challenges & Limitations

- Postal Code coverage: only ~20% of rows had a populated postal code, which is why the column was dropped rather than used for finer-grained geographic analysis.
- Currency assumption: the Sales, Profit, and Shipping Cost fields are treated as a single currency (USD) across all markets; the source dataset does not document per-market currency conversion, so true local-currency effects are not modeled.
- Surrogate key generation: LocationKey and DateKey were generated with a simple incrementing range() rather than a database identity/sequence, which is appropriate for a one-time batch build but would need to be revisited for an incrementally refreshed warehouse.
- Negative profit rows (23.9% of the data) were retained as legitimate loss-making orders rather than treated as errors; this is a modeling choice that should be re-confirmed against real business context if this were a production dataset.
- Static export pipeline: the current ETL process (CSV → CSV) is a manual, notebook-driven batch process; a production version of this warehouse would benefit from a scheduled ETL/ELT pipeline rather than manual notebook execution.

12. Conclusion

This project carried the Global Superstore dataset through a complete, end-to-end analytics lifecycle: from a raw, uncleaned 51,290-row extract to a validated dataset, a dimensional star-schema warehouse, a structured SQL analysis layer, and a five-page interactive Power BI report. Each stage was built to directly support the next — the cleaning notebook's feature engineering fed the warehouse build, the warehouse's fact and dimension tables fed the SQL queries, and the SQL queries' logic was mirrored directly into the Power BI visuals.

The resulting analysis identifies concrete, decision-relevant patterns — the profit impact of heavy discounting, the outsized profitability of the Technology category, the concentration of value in a small group of top customers, and the elevated shipping costs in Asian regions — that would give a retail business clear, prioritized places to focus its next round of pricing, product, and logistics decisions.

Beyond the specific findings, the project demonstrates a repeatable analytics workflow — clean → model → query → visualize — that can be applied to future data extracts from the same business with minimal rework, since the warehouse schema and dashboard structure are already in place.

Appendix A — Full Data Dictionary (Cleaned Dataset)

Column	Type	Notes
Row ID	Integer	Unique line-item identifier
Order ID	Text	Order identifier; repeats across line items in the same order
Order Date / Ship Date	Date	Converted to datetime64 during cleaning
Ship Mode	Text	4 categories
Customer ID / Name / Segment	Text	1,590 unique customers
City / State / Country / Market / Region	Text	Geographic hierarchy
Product ID / Name / Category / Sub-Category	Text	10,292 unique products across 3 categories, 17 sub-categories
Sales / Quantity / Discount / Profit / Shipping Cost	Numeric	Core transactional measures
Order Priority	Text	Low / Medium / High / Critical
Order Year / Order Month	Engineered	Derived from Order Date
Shipping Days	Engineered	Ship Date − Order Date, in days
Discount Level	Engineered	No Discount / Low / Medium / High buckets
Profit Category	Engineered	Loss / Low / Medium / High buckets

Appendix B — Project File Inventory

File	Description
Global_Superstore2.csv	Original raw dataset (51,290 rows, 24 columns)
final_project_clean.ipynb	Data cleaning, validation, feature engineering, and EDA notebook
final_project_warehouse.ipynb	Star-schema warehouse build notebook (dimension + fact table export)
Executive_Overview_dash1.sql	SQL queries and insights for Dashboard 1
Product_Performance_dash2.sql	SQL queries and insights for Dashboard 2
Customer___Geographic_Analysis_dash3.sql	SQL queries and insights for Dashboard 3
finall.pbix	Power BI Desktop report file (5 pages)